Rates and Unit Rates

A unit rate is how much for 1.

12 apples cost $6. How much for 1?

apples	12	6	2	1
dollars	6	3	1	0.5

Halve both, and halve again, until there is 1 apple.
$6 ÷ 12 = $0.50 each.

Per means “for each 1”

150 miles in 3 hours = 50 miles per hour

150 ÷ 3 = 50. Then 5 hours is 5 × 50 = 250 miles.

The better buy has the lower price for 1

4 pens for $2 is $0.50 each. 6 pens for $2.40 is $0.40 each: the better buy.

Your turn!

96 words in 4 minutes = 24 words per minute

For teachers

Goal: unit rates and using them (6.RP.2, 6.RP.3b). Say “How much for 1? Then multiply for any number.”
Prices in money: divide the cents.

Rates and Unit Rates: Practice

Find 1 first. Then multiply.

Part 1 Find 1 in the table Divide down to 1, then multiply.

1)

tickets	4	1	3	6
dollars	32	8	24	48

2)

hours	3	1	2	5
miles	165	55	110	275

3)

boxes	5	1	4	7
cookies	60	12	48	84

4)

days	6	1	2	9
pages	90	15	30	135

5)

bags	2	1	5	8
apples	14	7	35	56

Part 2 Unit rates No help this time.

6) 120 miles in 3 hours = 40 miles per hour

7) 84 words in 4 minutes = 21 words per minute

Rates and Unit Rates: Practice (continued)

Part 2 Unit rates (continued)

8) 90 pages in 5 days = 18 pages per day

9) 240 miles in 6 hours = 40 miles per hour

10) 56 laps in 8 hours = 7 laps per hour

11) 135 points in 9 games = 15 points per game

12) 72 cookies in 3 batches = 24 cookies per batch

13) 325 miles in 5 hours = 65 miles per hour

14) 96 dollars in 12 hours = 8 dollars per hour

15) 63 beads in 7 bracelets = 9 beads per bracelet

16) 180 steps in 4 minutes = 45 steps per minute

17) 144 dollars in 6 days = 24 dollars per day

18) 85 words in 5 minutes = 17 words per minute

19) 210 miles in 7 hours = 30 miles per hour

Rates and Unit Rates: Practice (continued)

Part 2 Unit rates (continued)

20) 48 eggs in 4 boxes = 12 eggs per box

21) 156 points in 3 games = 52 points per game

Part 3 Price for 1 No help this time.

22) 6 pencils for $3.00 = $0.50 each

23) 4 apples for $3.00 = $0.75 each

24) 8 stickers for $2.00 = $0.25 each

25) 5 notebooks for $6.00 = $1.20 each

26) 3 pears for $1.35 = $0.45 each

27) 10 erasers for $3.50 = $0.35 each

28) 12 candies for $1.80 = $0.15 each

29) 4 juices for $5.20 = $1.30 each

30) 6 bagels for $5.10 = $0.85 each

31) 5 cards for $3.00 = $0.60 each

Rates and Unit Rates: Practice (continued)

Part 3 Price for 1 (continued)

32) 8 pens for $3.20 = $0.40 each

33) 3 books for $6.75 = $2.25 each

Part 4 Use the rate No help this time.

34) 45 miles per hour: 6 hours = 270 miles

35) 12 pages per day: 9 days = 108 pages

36) 8 dollars per hour: 15 hours = 120 dollars

37) 65 words per minute: 4 minutes = 260 words

38) 24 cookies per batch: 7 batches = 168 cookies

39) 15 laps per hour: 8 hours = 120 laps

40) 35 miles per hour: 12 hours = 420 miles

41) 9 beads per bracelet: 14 bracelets = 126 beads

42) 18 points per game: 11 games = 198 points

43) 55 steps per minute: 6 minutes = 330 steps

Rates and Unit Rates: Practice (continued)

Part 4 Use the rate (continued)

44) 14 dollars per day: 21 days = 294 dollars

45) 6 eggs per box: 25 boxes = 150 eggs

Part 5 More unit rates

46) 270 pages in 5 days = 54 pages per day

47) 105 pages in 7 days = 15 pages per day

48) 44 words in 4 minutes = 11 words per minute

49) 504 words in 9 minutes = 56 words per minute

50) 18 words in 2 minutes = 9 words per minute

51) 112 dollars in 2 hours = 56 dollars per hour

52) 140 dollars in 5 hours = 28 dollars per hour

53) 354 words in 6 minutes = 59 words per minute

54) 153 points in 3 games = 51 points per game

Rates and Unit Rates: Practice (continued)

Part 5 More unit rates (continued)

55) 531 words in 9 minutes = 59 words per minute

56) 77 pages in 7 days = 11 pages per day

57) 192 miles in 3 hours = 64 miles per hour

58) 90 dollars in 2 hours = 45 dollars per hour

59) 120 pages in 8 days = 15 pages per day

60) 213 miles in 3 hours = 71 miles per hour

61) 130 words in 5 minutes = 26 words per minute

62) 360 words in 8 minutes = 45 words per minute

63) 54 pages in 6 days = 9 pages per day

Part 6 More prices for 1

64) 9 bagels for $25.20 = $2.80 each

65) 3 pens for $2.55 = $0.85 each

Rates and Unit Rates: Practice (continued)

Part 6 More prices for 1 (continued)

66) 9 cards for $11.25 = $1.25 each

67) 12 bagels for $8.40 = $0.70 each

68) 5 pens for $4.25 = $0.85 each

69) 9 juices for $4.50 = $0.50 each

70) 8 cards for $1.20 = $0.15 each

71) 2 erasers for $1.80 = $0.90 each

72) 4 cards for $2.60 = $0.65 each

73) 6 bagels for $2.70 = $0.45 each

74) 5 bagels for $3.00 = $0.60 each

75) 2 juices for $4.60 = $2.30 each

Rates and Unit Rates: Practice (continued)

Part 7 Remember? Problems from earlier packets.

76) $11 : 4 = \boxed{44} : 16$

77) $2.4 + 3.65 = \boxed{6.05}$

78) $\frac{1}{2} \times \frac{3}{5} = \frac{\boxed{3}}{\boxed{10}}$

79) $6 \times 2.35 = \boxed{14.1}$

80) $5.6 \div 7 = \boxed{0.8}$

81) $12 + 6 \times 3 = \boxed{30}$

Part 8 Word problems Find 1 first.

82) A car goes 270 miles in 6 hours. How far does it go in 1 hour?

Number sentence 270 ÷ 6 = 45

Answer 45 miles

Draw or show your work

83) A pack of 8 juice boxes costs $4. How much is 1 juice box?

Number sentence $4 ÷ 8 = $0.50

Answer $0.50

Draw or show your work

84) Mia types 42 words a minute. How many words can she type in 9 minutes?

Number sentence 42 × 9 = 378

Answer 378 words

Draw or show your work

Rates and Unit Rates: Practice (continued)

Part 8 Word problems (continued)

85) Store A: 5 pens for $3. Store B: 4 pens for $2. Which store has the lower price for 1 pen?

Draw or show your work

Number sentence $3 ÷ 5 = $0.60, $2 ÷ 4 = $0.50

Answer **Store B**

86) Challenge: A printer prints 36 pages in 3 minutes. How long does it take to print 180 pages?

Draw or show your work

Number sentence 36 ÷ 3 = 12 a minute, 180 ÷ 12 = 15

Answer **15** minutes

Check-In: Rates and Unit Rates

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Find 1 in the table

1)

boxes	3	1	2	7
pens	36	12	24	84

Part 2 Unit rates

2) 64 words in 4 minutes = 16 words per minute

3) 108 points in 9 games = 12 points per game

4) 98 dollars in 7 days = 14 dollars per day

Part 3 Price for 1

5) 5 pens for $2.00 = $0.40 each

6) 6 cards for $1.80 = $0.30 each

Part 4 Use the rate

7) 55 miles per hour: 4 hours = 220 miles

8) 16 pages per day: 7 days = 112 pages